

DR. SHAKUNTALA MISRA NATIONAL REHABILITATION UNIVERSITY
FACULTY OF ENGINEERING TECHNOLOGY,
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

STUDY & EVALUATION SCHEME

B.Tech. (CSE)

II Year: III Semester

II Year: III Semester											
S.No.	Paper Code	Paper Name	L	T	P	Sessional Assessment			SEE	Paper Total	Credit
						MST	TA	Total			
THEORY SUBJECT											
1	TCS301	Soft Computing	2	0	0	20	10	30	70	100	2
2	TCS302	Web Technology	3	0	0	20	10	30	70	100	3
3	TCS303	Data Structures & Algorithms	3	1	0	20	10	30	70	100	4
4	TES301	Digital Electronics	3	0	0	20	10	30	70	100	3
5	THS302	Industrial Sociology	2	0	0	20	10	30	70	100	2
6	TCS304	Discrete Mathematics	3	1	0	20	10	30	70	100	4
PRACTICAL/DESIGN/DRAWING											
7	TCS351	Soft Computing Lab	0	0	2	0	30	30	70	100	1
8	TCS352	Web Technology Lab	0	0	2	0	30	30	70	100	1
9	TCS 353	Data Structures Lab	0	0	2	0	30	30	70	100	1
10	TES351	Digital Electronics Lab	0	0	2	0	30	30	70	100	1
		Total	16	2	8					1000	22

L- Lecture

T -Tutorial

P-Practical

MST- Mid Semester Test

TA-Teacher's Assessment

SEE- Semester End Examination

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DIGITAL ELECTRONICS LAB

(TES351)

List of Experiments

1. To verify the truth tables of basic logic gates: AND, OR, NOR, NAND. Also to verify the truth table of Ex-OR, Ex-NOR (For 2, 3, & 4 inputs using gates).
2. To realization of OR, AND, NOR, Ex-OR, Ex-NOR using NAND & NOR gates.
3. To realize an SOP and POS expression.
4. To realize Half adder/ Subtractor & Full Adder/ Subtractor using NAND & NOR gates and to verify their truth tables
5. To realize a 4-bit ripple adder/ Subtractor using basic Half adder/ Subtractor & basic Full Adder/ Subtractor.
6. To verify the truth table of 4-to-1 multiplexer and 1-to-4 multiplexer. Realize the multiplexer using basic gates only. Also to construct an 8-to-1 multiplexer and 1-to-8 demultiplexer using blocks of 4-to-1 multiplexer and 1-to-4 demultiplexer
7. Design & Realize a combinational circuit that will accept a 2421 BCD code and drive a TIL -312 seven-segment display
8. Using basic logic gates, realize the R-S, J-K and D-flip flops with and without clock signal and verify their truth table.
9. Construct a divide by 2, 4 & 8 asynchronous counter. Construct a 4-bit binary counter and ring counter for a particular output pattern using D flip flop.
10. Perform input/output operations on parallel in/Parallel out and Serial in/Serial out registers using clock. Also exercise loading only one of multiple values into the register using multiplexer.

Note: As far as possible, the experiments shall be performed on bread board. However experiment Nos. 1-4 are to be performed on bread board only

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DATA STRUCTURE USING C LAB (TCS353)

LIST OF EXPERIMENTS

1. Sorting Algorithms-Non-Recursive.
2. Sorting Algorithms-Recursive.
3. Searching Algorithm.
4. Implementation of Stack using Array.
5. Implementation of Queue using Array.
6. Implementation of Circular Queue using Array.
7. Implementation of Stack using Linked List.
8. Implementation of Queue using Linked List.
9. Implementation of Circular Queue using Linked List.
10. Implementation of Tree Structures, Binary Tree, Tree Traversal, Binary Search Tree, Insertion and Deletion in BST.
11. Graph Implementation, BFS, DFS, Minimum cost spanning tree, shortest path algorithm.

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WEB TECHNOLOGY LAB
(TCS352)

LIST OF EXPERIMENTS

1. Write a program in HTML to display different styles of heading text.
2. Write a program to display the processes to be followed for a patient when he enters for a complete check-up. Use ordered lists and unordered lists.
3. Write a program to display a traditional Newspaper with the use of table tags.
4. With the help of "IMAGE" tags write a program to display the image along with some contents.
5. Use "Anchor" tag to write a program for displaying various Menus.
6. Use mapping technique, to map a particular part of image and move the control corresponding to that area. For e.g., in an image, if there are bat, ball, stump etc. When you click stump control should move to a file call St.htm.
7. Create frames that have details about various cities.
8. Create a form to display the kinds of food available in a Restaurant. (Use checkboxes wherever necessary)
9. Write a program to "reload" a page automatically once in 5 seconds.
10. Write a program using CSS to set the background colour, font, and paragraph.

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SOFT COMPUTING LAB (TCS351)

LIST OF EXPERIMENTS

1. Create a perceptron with appropriate no. of inputs and outputs. Train it using fixed increment learning algorithm until no change in weights is required. Output the final weights.
2. Create a simple ADALINE network with appropriate no. of input and output nodes. Train it using delta learning rule until no change in weights is required. Output the final weights.
3. Train the autocorrelator by given patterns: $A1=(-1,1,-1,1)$, $A2=(1,1,1,-1)$, $A3=(-1,-1,-1,1)$. Test it using patterns: $Ax=(-1,1,-1,1)$, $Ay=(1,1,1,1)$, $Az=(-1,-1,-1,-1)$.
4. Train the hetrocorrelator using multiple training encoding strategy for given patterns: $A1=(000111001)$ $B1=(010000111)$, $A2=(111001110)$ $B2=(100000001)$, $A3=(110110101)$ $B3(101001010)$. Test it using pattern A2
5. Implement Union, Intersection, Complement and Difference operations on fuzzy sets. Also create fuzzy relation by Cartesian product of any two fuzzy sets and perform maxmin composition on any two fuzzy relations.
6. Solve Greg Viot's fuzzy cruise controller using MATLAB Fuzzy logic toolbox.
7. Solve Air Conditioner Controller using MATLAB Fuzzy logic toolbox.
8. Implement TSP using GA.

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DISCRETE MATHEMATICS (TCS304)

Unit	Topic
I	Sets, relations and functions: Basic operations on sets, Cartesian products, disjoint union (sum), and power sets. Different types of relations, their compositions and inverses. Different types of functions, their compositions and inverses.
II	Propositional Logic: Syntax and semantics, proof systems, satisfiability, validity, soundness, completeness, deduction theorem, etc. Decision problems of propositional logic. Introduction to first order logic and first order theory. Partially ordered sets: Complete partial ordering, chain, lattice, complete, distributive, modular and complemented lattices. Boolean and pseudo Boolean lattices.
III	Algebraic Structures: Algebraic structures with one binary operation - semigroup, monoid and group. Cosets, Lagrange's theorem, normal subgroup, homomorphic subgroup. Congruence relation and quotient structures. Error correcting code. Algebraic structures with two binary operations- ring, integral domain, and field. Boolean algebra and boolean ring (Definitions and simple examples only).
IV	Introduction to Counting: Basic counting techniques - inclusion and exclusion, pigeon-hole principle, permutation, combination, summations. Introduction to recurrence relation and generating functions.
V	Introduction to Graphs: Graphs and their basic properties - degree, path, cycle, subgraph, isomorphism, Eulerian and Hamiltonian walk, trees.

Text Book (s):

1. C. L. Liu, Elements of Discrete Mathematics, 2nd Ed., Tata McGraw-Hill.
2. R. C. Penner, Discrete Mathematics: Proof Techniques and Mathematical Structures, World Scientific.
3. T. Veerarajan, "Discrete Mathematics", Tata McGraw-Hill
4. K. H. Rosen, Discrete Mathematics and its Applications, 6th Ed., Tata McGraw-Hill.
5. J. L. Hein, Discrete Structures, Logic, and Computability, 3rd Ed., Jones and Bartlett.
6. N. Deo, Graph Theory, Prentice Hall of India.
7. S. Lipschutz and M. L. Lipson, Schaum's Outline of Theory and Problems of Discrete Mathematics, 2nd Ed., Tata McGraw-Hill.
8. J. P. Tremblay and R. P. Manohar, Discrete Mathematics with Applications to Computer Science, Tata McGraw-Hill.

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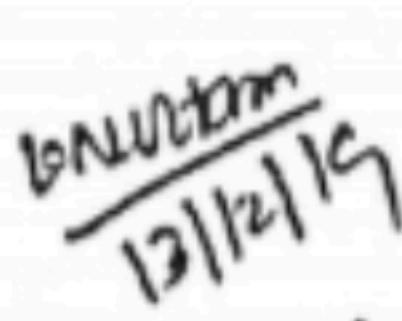
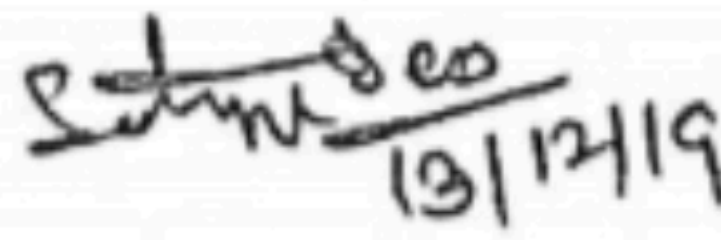
Industrial Sociology
Course Code: THS- 302/THS-402

Unit	Contents	No. of Hrs.
I	Descriptors/Topics Industrial Sociology: Nature, Scope and Importance, Origin and Development, Industry as a social system, Evolution of Working Class , Changing nature of work , Growth of unorganized informal sector. Dynamics of Industrial Relations: Approaches to the study of Industrial Relations, Collective Bargaining – Concepts, Types, Scope and Importance.	12
II	Industrial Disputes: Concept, Features and Kinds of disputes, Settling disputes, Mediation, Arbitration, Conciliation, Negotiation.	8
III	Descriptors /Topics Trade Union: Concept, Features, Functions and Types, History of Trade Union Movement in India, Trade Unions and Challenges of Privatization and Globalization, Law and work, Decline of Trade Unions.	10
IV	Dynamics of Industrial Relations: Corporate Social Responsibility, Inclusion of Women in the Corporate Sector, Scope of Industrial Sociology in India, Impact on Employment, Impact on HRD, Impact on wages and benefits, Modern Industry in India – Development of Industry in Post-Independence period, The Indian Worker: Features of Indian worker, The contribution of Social - Philosophy, family, caste and community in determining the attitude of workers.	10

References:

1. PREMVAR KAPOOR, Sociology & Economics for Engineers, Khanna Publishing House (Edition 2018).
2. GIBERT PASCAL, Fundamentals of Industrial sociology, Tata McGraw Hill, New Delhi, 1972.
3. MAMORIA C.B. And MAMORIA S., Dynamics of Industrial Relations in India.
4. SINHA G.P. and P.R.N. SINHA, Industrial Relations and Labour Legislations, New Delhi, Oxford and IBH Publishing Co., 1977.
5. S.C. SHARMA, Industrial Safety and Health Management, Khanna Book Publishing Co. (P) Ltd., Delhi (ISBN: 978-93-86173-188



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 13/12/19
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DIGITAL ELECTRONICS (TES301)

Unit	Topic
I	Fundamentals of Digital Systems and logic families: Digital signals, digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, Boolean algebra, examples of IC gates, number systems-binary, signed binary, octal, hexadecimal number, binary arithmetic, one's and two's complements arithmetic, codes, error detecting and correcting codes, characteristics of digital ICs, digital logic families, TTL, Schottky TTL, and CMOS logic, interfacing CMOS and TTL, Tri-state logic.
II	Combinational Digital Circuits Standard representation for logic functions, K-map representation, simplification of logic functions using K-map, minimization of logical functions. Don't care conditions, Multiplexer, De- Multiplexer/Decoders, Adders, Subtractors, BCD arithmetic, carry look ahead adder, serial adder, ALU, elementary ALU design, popular MSI chips, digital comparator, parity checker/generator, code converters, priority encoders, decoders/drivers for display devices, Q-M method of function realization.
III	Sequential circuits and systems A 1-bit memory, the circuit properties of Bi stable latch, the clocked SR flip flop, J- K-T and D types flip flops, applications of flip flops, shift registers, applications of shift registers, serial to parallel converter, parallel to serial converter, ring counter, sequence generator, ripple (Asynchronous) counters, synchronous counters, counters design using flip flops, special counter IC's, asynchronous sequential counters, applications of counters..
IV	A/D and D/A Converters: Digital to analog converters: weighted resistor/converter, R-2R Ladder D/A converter, specifications for D/A converters, examples of D/A converter ICs, sample and hold circuit, analog to digital converters: quantization and encoding, parallel comparator A/D converter, successive approximation A/D converter, counting A/D converter, dual slope A/D converter, A/D converter using voltage to frequency and voltage to time conversion, specifications of A/D converters, example of A/D converter ICs
V	Semiconductor memories and Programmable logic devices. Memory organization and operation, expanding memory size, classification and characteristics of memories, sequential memory, read only memory (ROM), read and write memory (RAM), content addressable memory (CAM), charge coupled device memory (CCD), commonly used memory chips, ROM as a PLD, Programmable logic array, Programmable array logic, complex Programmable logic devices (CPLDS), Field Programmable Gate Array (FPGA)..

Text Book (s):

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
3. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016..



DATA STRUCTURES & ALGORITHMS (TCS303)

Unit	Topic
I	Introduction: Basic Terminologies: Elementary Data Organizations, Data Structure Operations: insertion, deletion, traversal etc.; Analysis of an Algorithm, Asymptotic Notations, Time-Space trade off. Searching: Linear Search and Binary Search Techniques and their complexity analysis.
II	Stacks and Queues: ADT Stack and its operations: Algorithms and their complexity analysis, Applications of Stacks: Expression Conversion and evaluation – corresponding algorithms and complexity analysis. ADT queue, Types of Queue: Simple Queue, Circular Queue, Priority Queue; Operations on each types of Queues: Algorithms and their analysis.
III	Linked Lists: Singly linked lists: Representation in memory, Algorithms of several operations: Traversing, Searching, Insertion into, Deletion from linked list; Linked representation of Stack and Queue, Header nodes, Doubly linked list: operations on it and algorithmic analysis; Circular Linked Lists: all operations their algorithms and the complexity analysis.
IV	Trees: Basic Tree Terminologies, Different types of Trees: Binary Tree, Threaded Binary Tree, Binary Search Tree, AVL Tree; Tree operations on each of the trees and their algorithms with complexity analysis. Applications of Binary Trees. B Tree, B+ Tree: definitions, algorithms and analysis.
V	Sorting and Hashing: Objective and properties of different sorting algorithms: Selection Sort, Bubble Sort, Insertion Sort, Quick Sort, Merge Sort, Heap Sort; Performance and Comparison among all the methods, Hashing. Graph: Basic Terminologies and Representations, Graph search and traversal algorithms and complexity analysis.

Text Book (s):

1. "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni. Computer Science Press.
2. "Data Structures", RS Salaria, Khanna Publishing House
3. "How to Solve it by Computer", 2nd Impression by R.G. Dromey, Pearson Education.

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WEB TECHNOLOGY (TCS302)

Unit	Topic
I	Introduction to Web Technology: Internet, WWW, Web Browsers with suitable examples, Web Servers with suitable examples, URL, HTTP, MIME, Introduction to HTML& DHTML : Basic Syntax, HTML Document Structure, Text Formatting, Images, Lists, Links, Tables, Frames, Forms, Cascade Style Sheets : Levels Of Style Sheets, Specification Formats, Style Classes, Properties, Colors, Span and Div tags.
II	Introduction to Java Script: Overview of java Script, Syntactic characteristics, Primitives, Operator and Expression, control statements, Arrays, functions, errors in scripts, Document Object Model(DOM),Event driven computation, element access in Java script, The navigator Object, Dynamic Document with Java Script : Element positioning, Moving elements, Changing colors and fonts, Dynamic content, Locating the mouse Cursor, Slow movements of elements, Dragging and Dropping Elements.
III	Introduction to XML: Syntax of XML, Document Structure, Document type definition, Namespaces,XML Schemas, Document Object model, Presenting XML, Using XML Processors: DOM and SAX.
IV	Introduction to PHP: Overview of PHP, general server characteristics, Creating PHP Pages, Form handling, Data Base access with PHP & MySql. JSP & Servlets: Basics of JSP, Tags, Session Handling, Redirection and Basics of Servlet
V	Database Connectivity: Connection to various databases (MySQL, Oracle), Database connectivity in PHP, JSP, Other server languages, Basic SQL Queries and Statements.

Text Book (s):

1. Programming world wide web- Robert W.Sebesta , Pearson.
2. Beginners PHP, Apache, MY Sql, Web Development, by Michael Glass, Wrox.

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SOFT COMPUTING (TCS301)

Unit	Topic
I	Neural Networks-I(Introduction & Architecture): Neuron, Nerve structure and synapse, Artificial Neuron and its model, activation functions, Neural network architecture: single layer and multilayer feed forward networks, recurrent networks. Various learning techniques; perception and convergence rule, Auto-associative and hetro-associative memory.
II	Neural Networks-II (Back propogation networks): Architecture perceptron model, solution, single layer artificial neural network, multilayer perceptron model; back propogation learning methods, effect of learning rule co-efficient ;back propogation algorithm, factors affecting backpropagation training, applications.
III	Fuzzy Logic-I (Introduction) Basic concepts of fuzzy logic, Fuzzy sets and Crisp sets, Fuzzy set theory and operations, Properties of fuzzy sets, Fuzzy and Crisp relations, Fuzzy to Crisp conversion.
IV	Fuzzy Logic -II (Fuzzy Membership, Rules): Membership functions, inference in fuzzy logic, fuzzy if-then rules, Fuzzy implications and Fuzzy algorithms, Fuzzyfications & Defuzzificataions, Fuzzy Controller, Industrial applications.
V	Genetic Algorithm(GA) Basic concepts, working principle, procedures of GA, flow chart of GA, Genetic representations, (encoding) Initialization and selection, Genetic operators, Mutation, Generational Cycle, applications.

Text Book (s):

1. S. Rajsekaran & G.A. Vijayalakshmi Pai, "Neural Networks, Fuzzy Logic and Genetic Algorithm: Synthesis and Applications" Prentice Hall of India.
2. N.P. Padhy, "Artificial Intelligence and Intelligent Systems" Oxford University Press.

Reference Books:

3. Siman Haykin, "Neural Netowrks" Prentice Hall of India
4. Timothy J. Ross, "Fuzzy Logic with Engineering Applications" Wiley India.
5. Kumar Satish, "Neural Networks" Tata Mc Graw Hill

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